

Submission LINALG1-FINAL-SUB01

Question LINALG1-FINAL-Q01

scratch: show step

1. $\det = 6(6) - 4(4) = 20$
 2. Eliminate y , yielding $x=1$; substitute $\Rightarrow y=2$
 3. checks: $6(1)+4(2)=14$, $4(1)+6(2)=16$
- $\Rightarrow$ FINAL: $x=1$, $y=2$ (check units/interpretation)

Question LINALG1-FINAL-Q02

scratch: show step

1. $\det(A) = 2 \cdot 6 - 2 \cdot 2 = 8$
 2. $\det(A) \neq 0$, so A is invertible.
- $\Rightarrow$ FINAL: $\det=8$; invertible (check units/interpretation)

Question LINALG1-FINAL-Q03

scratch: show step

1. $T(1, -1) = (1 - 2, 3 + 1) = (-1, 4)$
 2. Matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$
 3. Both coordinate rules have no constant term, so T is linear.
- $\Rightarrow$ FINAL: $T(1, -1) = (-1, 4)$; matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$; linear (check units/interpretation)

Question LINALG1-FINAL-Q04

scratch: show step

1. Set $c \blacksquare (1, 0) + c \blacksquare (0, 1) = (0, 0)$.
 2. Coordinates give $c \blacksquare = 0$ and $c \blacksquare = 0$ only, so the set is independent.
- $\Rightarrow$ FINAL: only trivial combination; $\{(1, 0), (0, 1)\}$ is linearly independent (check units/interpretation)

Question LINALG1-FINAL-Q05

scratch: show step

1. $\text{Null}(A)$ is the set of vectors sent to the zero vector.
 2. It is the solution space of $Ax=0$ and is a subspace.
- $\Rightarrow$ FINAL: the vectors mapped to zero; exactly the solution space of $Ax=0$ (check units/interpretation)

Question LINALG1-FINAL-Q06

scratch: show step

1. $\det = 6(6) - 1(4) = 32$
 2. Eliminate y , yielding $x=1$; substitute $\Rightarrow y=2$
 3. checks: $6(1)+1(2)=8$, $4(1)+6(2)=16$
- $\Rightarrow$ FINAL: $x=1$, $y=2$ (check units/interpretation)

Question LINALG1-FINAL-Q07

scratch: show step

1. $\det(A) = 4 \cdot 4 - 2 \cdot 2 = 12$

2. $\det(A) \neq 0$, so A is invertible.

=> FINAL: $\det=12$; invertible (check units/interpretation)

Question LINALG1-FINAL-Q08

scratch: show step

1. $T(1, -1) = (1 - 2, 3 + 1) = (-1, 4)$

2. Matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$

3. Both coordinate rules have no constant term, so T is linear.

=> FINAL: $T(1, -1) = (-1, 4)$; matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$; linear (check units/interpretation)

Question LINALG1-FINAL-Q09

scratch: show step

1. Set $c_1(1, 0) + c_2(0, 1) = (0, 0)$.

2. Coordinates give $c_1 = 0$ and $c_2 = 0$ only, so the set is independent.

=> FINAL: only trivial combination; $\{(1, 0), (0, 1)\}$ is linearly independent (check units/interpretation)

Question LINALG1-FINAL-Q10

scratch: show step

1. $\text{Null}(A)$ is the set of vectors sent to the zero vector.

2. It is the solution space of $Ax=0$ and is a subspace.

=> FINAL: the vectors mapped to zero; exactly the solution space of $Ax=0$ (check units/interpretation)